$$\textit{Find the tangent line } g(x)=\frac{16}{x}-4\sqrt{x} \textit{ at } x=4$$

$$g(x)=16x^{-1}-4x^{\frac{1}{2}}$$

$$\frac{dg}{dx}=-16x^{-2}-2x^{\frac{1}{2}-\frac{2}{2}}$$

$$\frac{dg}{dx}=-16x^{-2}-2x^{-\frac{1}{2}}$$

$$\frac{dg}{dx}(4)=-\frac{16}{4^2}-\frac{2}{\sqrt{4}}$$

$$\frac{dg}{dx}(4)=-1-1=-2$$

$$m=-2$$

$$y-y_1=m(x-x_1)$$

$$y_1=\frac{16}{4}-4\sqrt{4}=4-4(2)=4-8=-4$$

$$y+4=-2(x-4)$$

$$y=-2x+8-4$$

$$y=-2x+4$$